

RAJASEKAR A

7 years of experience in Embedded System Design and Development

LinkedIn: <https://www.linkedin.com/in/rajasekar-selva-699bb6b7/>

BANASHANKARI, BANGALORE

+91- 6383828407

selvarajasekar96@gmail.com

RESUME PROFILE

- Seasoned Senior Embedded Software Developer with 7 years of experience
- Skilled in Embedded C programming and various microcontroller platforms.
- Familiar with version control systems like GitHub and Bit bucket.
- Proficient in utilizing various microcontroller types, including 8051, Microchip, STM32 ARM Cortex-M series, Renesas, NXP Semiconductor S32K3 series, Dialog Semiconductor DA1470x series, to facilitate efficient development across diverse platforms.
- Seeking a challenging role in firmware development and embedded systems.
- Proficient in various IDEs and simulation software, adept at navigating different development environments with ease.
- Ability to interact with Testing Team and product analysts regarding Project status.
- Strong problem-solving abilities and system debugging skills.

WORK EXPERIENCE

Claysol Media Labs Private Limited, Bangalore

OCTOBER 2023 - PRESENT

ROLE: Senior Embedded Software Developer

1.PROJECT— *PMV ELECTRIC VEHICLE*

Description: The Eco-Friendly and Sustainable - Electric (EaS-E®), is a fully electric smart car for everyday use. Our motto is towards achieving sustainable long-term personal mobility to address the global challenges of pollution, climate change, urban congestion and even more recent challenges related to parking. The purpose built Personal Mobility Vehicle (PMV®) is the future of urban mobility for intra-city everyday commute.

Performed Role:

Orchestrated the firmware architecture for a vehicle control unit, incorporating peripheral devices like sensors, actuators, and communication interfaces such as UART, SPI, I2C, and CAN.

Implemented firmware utilizing the real-time operating system FreeRTOS, ensuring efficient task management and resource utilization.

Developed the application layer specific to the vehicle control unit, enabling comprehensive control and management of vehicle functionalities.

Integrated inputs and outputs within the firmware to facilitate seamless communication between sensors, actuators, and the VCU's processing unit.

Skills

Programming languages:

C, Embedded C, Python.

Controllers

Architecture:

Harvard, ARM Cortex M33 Cortex M0+, Cortex M3, Renesas RL Series, Cortex M7.

Series:

NXP -
S32K3xx, S32K1xx,
LPC214x
Dialog Semiconductor-
DA1470x
MICROCHIP-
PIC16F877A,
PIC32MX795F512,
Renesas -
RL78X, RA2, RA6 series
MCS51-80C51
STMicroelectronics -
STM32LX series,
STM32FX series

Protocols:

UART, SPI, I2C, CAN and MODBUS.

MISCELLANEOUS:

PWM, Timers (Software and Hardware).
ISR and Linker Scripts.

OS: FREERTOS

INDUSTRY: Electric Vehicles, Medical Devices and Industrial Automation.

IDE

S32 DESIGN STUDIO

Controller(PMV)

S32K312

Tools

PE micro debugger,
PCAN-view, CAN db++,
Proteus,
J-Link, ST-Link, JTAG,
and SWD

Technosphere Labs Private Limited, Bangalore

APRIL 2022 – OCTOBER 2023

ROLE: Embedded Software Developer

1.PROJECT— *TJAY*

Description: TJay is designed for the early detection and management of epilepsy. With TJay, we hope to make the lives of people with epilepsy safer and remove some of the uncertainties they deal with every day. We are also trying to make it easier for doctors to arrive at a more accurate and faster diagnosis, with the help of streaming real-time data, collected over long periods of time.

Performed Role:

Designed the firmware architecture for the wrist section of TJay, integrating various communication protocols such as SPI, I2C, UART, and Bluetooth.

Integrated sensors including BMI270, BMM150, MAX30208, MAX86141 and a temperature sensor.

Implemented queue management for multitasking, circular buffer for data storage, and RTC for timestamping.

Developed the firmware utilizing the real-time operating system FreeRTOS and implemented the application layer for the wrist section.

2.PROJECT— *COMMUNICATION INTERFACE BOARD*

Description: The primary goal of the CIB project is to install CIB boards in various locations within metro stations, particularly in lifts. These boards are designed to monitor and gather data from the lifts, leveraging CAN communication for seamless integration and enhancing operational efficiency and safety in metro stations.

Performed Role:

Designed the firmware architecture for the CIB, developing it on a bare-metal platform.

Integrated peripheral devices including MODBUS, SPI, and CAN, ensuring seamless communication within the system.

Developed the application layer for the CIB, enabling functions such as data collection and storage onto an SD card.

3.PROJECT— *BLDC FAN*

Description: The objective of the BLDC fan project is to advocate for energy efficiency and sustainability through the adoption of Brushless DC (BLDC) technology in fan systems.

Performed Role:

Transferred code from Microchip to STM platform, ensuring compatibility and functionality across different hardware architectures.

Optimized memory usage during the porting process to maximize efficiency and resource utilization on the STM platform.

Developed firmware specifically for the application layer of BLDC systems, focusing on functionalities such as motor control and system management.

Conducted comprehensive testing using MotorXpert to verify the performance and reliability of the BLDC application layer firmware on the STM platform.

Controller

DA14706

Multicore processor

Protocol

RS485

CAN

Sensors

BMM150(Magnetometer)

BMI270(Accelerometer

&Gyroscope)

MAX30208(Temperature sensor)

MAX86140(PPG)

MAX30001(EDA)

Application

Sensor logger

NRF Connect

Script

Python Script

IDE

Smart snippets

Controller

PIC32MX795F12L

32bit PIC Microcontroller

MCP2515

Protocol

SPI

I2C

UART

BLUETOOTH

IDE

MPLAB X IDE

Flash Tool

MPLAB IPE

Controller

STM32G030K6

Protocol

RF

UART

IDE

STM32 CUBE IDE

Flash Tool

ST-Link

Propel Industries Private Limited, Coimbatore

JANUARY 2021 – MARCH 2022

ROLE: Embedded Software Developer

1.PROJECT— *AVC POWER PACK*

Description: The AVC power pack is a central component in the cone crusher system, providing vital control and functionality. Using the MODBUS RTU protocol, it monitors sensor values for smooth operation. Additionally, the power pack manages the start and stop sequences, regulates heater and cooler functions, issues lub ready commands, and oversees critical tasks, ensuring the optimal performance and longevity of the equipment.

Performed Role:

Developed low-level drivers for UART and GLCD interfaces and created the application layer for the AVC power pack.

This layer contained logic to read analog sensor values and manage the parallel control of heater and cooler functions.

Additionally, it facilitated data transmission through MODBUS RTU protocol for effective communication within the system.

Conducted testing, and the project was developed in a bare-metal environment.

2.PROJECT— *ECOWASH*

Description: Propel Eco Wash provides reliable solutions for material washing, with precise silt and clay separation. Embedded sensors control motor operation and trigger heater/cooler activation. The GLCD displays real-time temperature and pressure readings with an RTC. Data logging via SPI onto an SD card enables comprehensive machine status tracking.

Performed Role:

Developed low-level drivers for UART and GLCD interfaces, along with an application layer for the ECO-WASH.

This layer included logic for reading analog sensor values and controlling heater and cooler functions in parallel. Furthermore, it enabled data logging to an SD card via SPI communication.

After testing and debugging, the project was completed in a bare-metal environment.

Zeno Research and Development, Thanjavur

FEBRUARY 2016 – NOVEMBER 2020

ROLE: Junior Embedded Software Developer

1.PROJECT— *Voice controlled Robot using Microchip*

Description This project introduces a voice-controlled robotic system that utilizes Bluetooth technology to execute human commands.

Voice commands are transmitted through an Android app via nRF Connect to the Bluetooth module, which interfaces with the controller.

Performed Role:

Developed low-level firmware for UART and BLUETOOTH.

Developed firmware for the application layer.

Tested and debugged using Proteus simulation.

Controller

RL78/G14

16bit Microcontroller

Protocol

MODBUS

UART

IDE

E2 STUDIO

Sensors

PT100(Temperature Sensor)

4-20ma (Pressure Sensor)

Flash Tool

Renesas Flash Programmer

Controller

RL78/G14

16bit Microcontroller

Protocol

SPI

UART

IDE

E2 STUDIO

Sensors

PT100(Temperature Sensor)

4-20ma (Pressure Sensor)

Flash Tool

Renesas Flash Programmer

Controller

PIC16F877A

8bit Microcontroller

Protocol

BLUETOOTH

UART

IDE

MPLAB X IDE

Flash Tool

MPLAB IPE

Application

NRF Connect

Simulation

Proteus

2.PROJECT— *Smart Meter using IOT*

Description: This project helps save energy at home by monitoring usage. It uses an Arduino-based wireless meter to measure electricity without invading wires. The data is sent over Wi-Fi to show how much energy is being used. This helps users understand and cut down on their energy consumption.

Performed Role:

Developed low-level firmware for UART and the Wi-Fi module (ESP8266).

Created low-level firmware for the 16x2 LCD display.

Integrated the current sensor (ACS712) and RTC into the system.

Controller

PIC16F877A

8bit Microcontroller

IDE

MPLAB X IDE

Simulation

Proteus

Web Application

Thingspeak

3.PROJECT— *Electronic Voting system using Finger print*

Description: Elections are vital in a democratic country, ensuring every citizen's right to vote. Government involvement is crucial for a successful election. This paper proposes a biometric verification system to enhance election security and reduce manpower needed, addressing current challenges with electronic voting machines.

Performed Role:

Integrated the finger print sensor and RTC in to the system.

Developed application-level drivers.

4.PROJECT— *Real Time GPS Tracking System*

Description: This paper presents a robust vehicle monitoring and tracking system using an ARM7 LPC2148 processor with MEMS, temperature sensors, GSM/GPRS, and GPS technologies. It ensures accurate location tracking and monitoring of vehicle parameters, enhancing safety and theft prevention.

Performed Role:

Interfaced GPS and GSM modules.

Interfaced MEMS sensor and LCD.

Developed application-level drivers.

5.PROJECT— *Propeller Display*

Description The Digital Propeller Clock with Android Override uses a Microchip microcontroller and a high RPM motor to display time in HH:MM:SS format via LEDs, leveraging persistence of vision. Users can synchronize and update the time using an Android device via Bluetooth, ensuring accurate and real-time display.

Performed Role:

Interfaced GPS and GSM modules.

Interfaced MEMS sensor and LCD.

Developed application-level drivers.

Controller

LPC2148

16bit Microcontroller

IDE

Keil

Simulation

Proteus

Controller

PIC16F877A

8bit Microcontroller

IDE

MPLAB X IDE

Simulation

Proteus

Application

NRF Connect

Educational Qualification:

I earned a Bachelor's degree in Electronics and Communication Engineering from Trichy Engineering College, Tamil Nadu, from 2011 to 2015.

Languages: English, Kannada and Tamil.